A robot that acts to know

Day 1 — The challenge, and free energy

Course built on the doctoral thesis of Noé Zapata

Monday 30 November

Montevideo, 30 November – 4 December 2026

The challenge

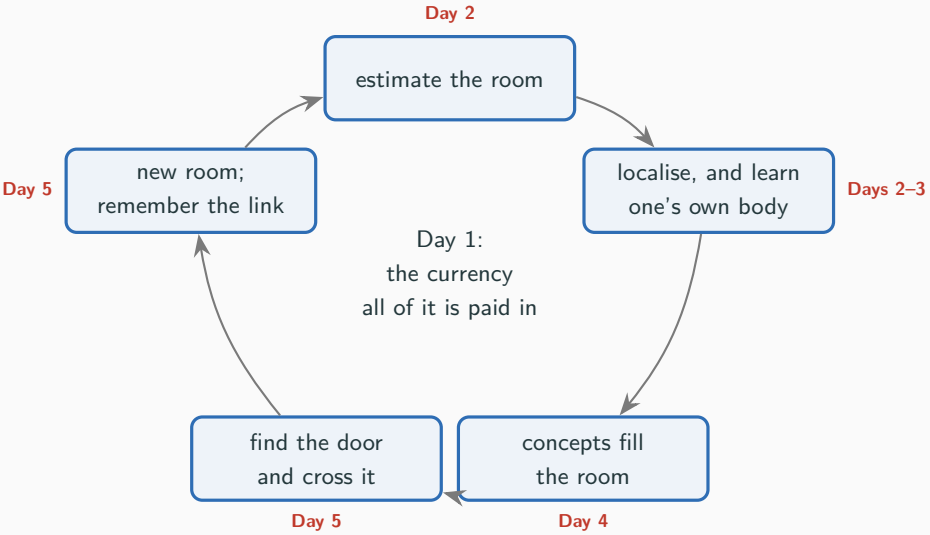
A robot is put down in an apartment it has never seen

[video: the full cycle, Webots / real apartment]

room estimated → localised → objects appear → door crossed → second room

Watch for one thing: nothing in this video was programmed as a sequence of steps.

The cycle we will build this week

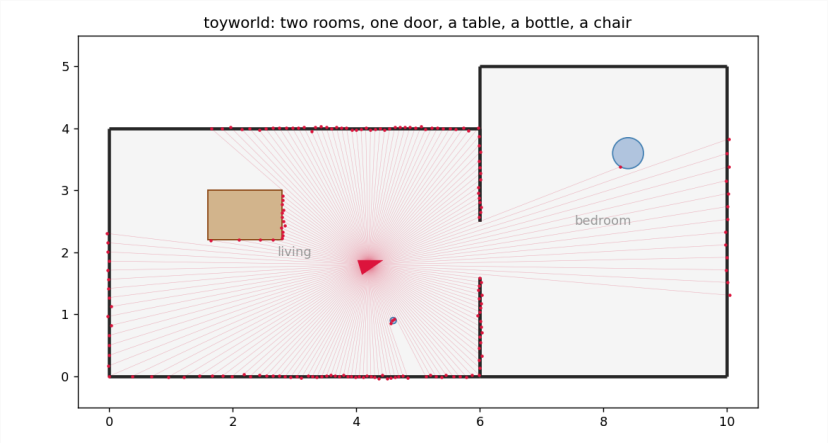


Every stage is a belief

stage	what is unknown (latent)	what supplies the evidence	day
estimate the room	walls, corners, the polygon	LiDAR returns	2
localise	the robot's pose in that room	LiDAR against the believed room	2
learn the body	wheel scale, gyro bias, sensor mounts	the localiser's corrections	3
concepts	which objects exist, where, what shape	camera masks, LiDAR	4
door	where it is, whether it is open	the room's boundary, free space	5
new room + memory	a second polygon; how rooms connect	all of the above, again	5

One generative model over all of it, and one quantity that every stage minimises: perception lowers it by changing beliefs, action by changing the data.

The toy version you will build, one stage a day



Two rooms, one door, three objects, a LiDAR, and a robot whose wheels lie a little. Pure `numpy`; it runs on your laptop.

How this course is built

Each session runs through three steps:

1. the **theory**, derived rather than asserted;
2. the **code**, one stage of the cycle, in the lab;
3. the **measurements** that showed when the equations were wrong.

The third step is not optional. Most of what the thesis learned, it learned there.

The thread through all five days

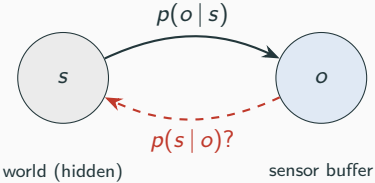
Averaging removes independent error.
It does not remove shared error.

You will meet it as:

- the common-mode ceiling;
- a prior re-centred on its own estimate;
- a fused precision better than its best member;
- a metric that scores what the robot never saw.

Inference behind a sensory boundary

The agent never sees the state



The **generative model**:

$$p(s, o) = p(o \mid s)\, p(s)$$

- $p(o \mid s)$ is the agent's theory of its own sensor: which LiDAR returns a robot at this pose, in this room, would produce.
- $p(s)$ is what it brought with it: plausible room shapes, typical table sizes.

Inference runs the arrow backwards:

$$p(s \mid o) = \frac{p(o \mid s)\, p(s)}{p(o)}$$

source: active_inference.tex l.181–212

Where exact inference dies

$$p(o) = \int p(o \mid s)\, p(s)\, \mathrm{d}s$$

- The probability of *this scan* under *every* room and furniture configuration the robot considers possible, each weighted by how likely it is.
- No closed form and no feasible quadrature. This is not a quirk of one sensor: it is the generic situation.
- Every practical method from here on is a way of not computing it.

A word about "state"

s is a latent variable of the *agent's model*, not the physical truth, which is never accessible. That distinction is what gives the question "is this belief wrong?" its meaning.

source: active_inference.tex l.200–220

Variational free energy

Do not compute the posterior; approximate it

Choose a tractable family, and a member $q(s)$ of it (the **belief**). Move it towards the true posterior, measuring closeness by

$$D_{\mathrm{KL}}[q(s) \,\|\, p(s \mid o)] = \int q(s) \ln \frac{q(s)}{p(s \mid o)} \,\mathrm{d}s \;\geq 0.$$

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Substitute Bayes' rule and separate out the part that does not depend on q:

$$D_{\mathrm{KL}}[q \,\|\, p(s \mid o)] = \underbrace{\int q(s) \ln \frac{q(s)}{p(o, s)} \,\mathrm{d}s}_{F[q,o]} \;+\; \ln p(o)$$

F needs only the belief and the joint $p(o \mid s)p(s)$ the agent already wrote down.

source: active_inference.tex l.222–250

The identity the framework rests on

$$F[q, o] = D_{\mathrm{KL}}[q(s) \,\|\, p(s \mid o)] - \ln p(o)$$

Reading 1: same minimiser

$\ln p(o)$ does not depend on q. Minimising the computable F minimises the uncomputable divergence.

Reading 2: a bound on surprise

$\mathrm{KL} \geq 0$, so $F \geq -\ln p(o)$. Lowering F makes the belief better *and* the data less surprising under the agent's own model.

If you know machine learning: $-F$ is the **ELBO**. Maximising the ELBO and minimising free energy are the same operation.

source: active_inference.tex l.252–268

Complexity minus accuracy

$$F[q, o] = \underbrace{D_{\mathrm{KL}}[q(s) \,\|\, p(s)]}_{\text{complexity}} - \underbrace{\mathbb{E}_q[\ln p(o \mid s)]}_{\text{accuracy}}$$

- **Accuracy** rewards beliefs under which the data are likely. On its own, it would adopt any explanation that fits the current scan.
- **Complexity** charges for the distance travelled from the prior: Occam's razor, written as a divergence.
- Free energy picks the belief that buys the most accuracy for the least departure from what was already believed.

One trade, stated once, in nats, for every quantity the agent believes.

source: active_inference.tex l.288–308

Worked example: the Kalman update is free-energy minimisation

Take the prior $s \sim \mathcal{N}(m, p)$, an observation $y \sim \mathcal{N}(s, r)$, and the belief $q = \mathcal{N}(\mu, \Sigma)$. Every term is a Gaussian expectation:

$$F(\mu, \Sigma) = \frac{(\mu - m)^2 + \Sigma}{2p} + \frac{(y - \mu)^2 + \Sigma}{2r} - \tfrac{1}{2} \ln \Sigma + \text{const}$$

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$\partial F / \partial \mu = 0$:

$$\mu = \frac{m/p + y/r}{1/p + 1/r}$$

the precision-weighted mean.

$\partial F / \partial \Sigma = 0$:

$$\frac{1}{\Sigma} = \frac{1}{p} + \frac{1}{r}$$

precisions add.

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The two halves come from different terms. The *mean* comes from the prediction errors; the *variance* comes from the curvature of F balanced against the entropy $-\frac{1}{2} \ln \Sigma$. Regularised least squares is the same trade with a flat prior.

A choice, marked as a choice

- Closeness was measured with KL. Nothing forced that: it is the $\alpha \to 1$ member of the Rényi family, and the bound survives the substitution.
- The order α controls the *character* of the error: does the belief err towards too broad, or towards too confident?
- On this robot, that is not a statistical nicety. A covariance is spent: on deciding where to look next, and on how far a measurement may move a belief.

The thesis keeps $\alpha \to 1$ throughout. Making α itself a latent variable is open future work.

source: active_inference.tex l.270–286; future_work.tex l.22

The Laplace approximation

Gaussian beliefs: the form everything later takes

Take $q(s) = \mathcal{N}(\mu, \Sigma)$ and expand around the mode:

$$F \approx -\ln p(o, \mu) + \tfrac{1}{2}\,\mathrm{tr}(\mathcal{H}\Sigma) - \tfrac{1}{2}\ln|\Sigma| - \tfrac{n}{2}\ln(2\pi e), \qquad \mathcal{H} = -\nabla_s^2 \ln p(o, s)\big|_{\mu}$$

- The **mean** moves to the mode of the joint: a nonlinear least-squares problem (Gauss–Newton).
- The **covariance** is pulled two ways: the curvature term shrinks it, the entropy term widens it. Setting $\partial F / \partial \Sigma = 0$:

$$\boxed{\Sigma = \mathcal{H}^{-1}}$$

The uncertainty comes as a by-product of fitting, at no extra modelling cost.

source: active_inference.tex l.310–333

Gauss–Newton, in one line

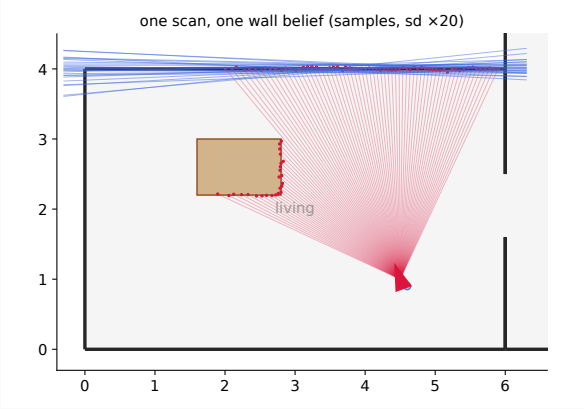
For whitened residuals $r(s)$ with noise covariance R, and a Gaussian prior $\mathcal{N}(m, P)$:

$$-\ln p(o, s) = \tfrac{1}{2} r^\top R^{-1} r + \tfrac{1}{2}(s-m)^\top P^{-1}(s-m) + \text{const},$$

$$\mathcal{H} \approx J^\top R^{-1} J + P^{-1}, \qquad \Delta s = -\mathcal{H}^{-1}\big(J^\top R^{-1} r + P^{-1}(s-m)\big).$$

- Iterate to convergence. The final $\mathcal{H}^{-1}$ *is* the belief's covariance.
- In the lab, `laplace(residual, x0, sigma, prior=...)` does exactly this. It also returns the Laplace free energy $F \approx -\ln p(o, \mu) + \frac{1}{2}\ln|\mathcal{H}| - \frac{n}{2}\ln 2\pi$, which you will use to compare models.

Example: one wall as a belief



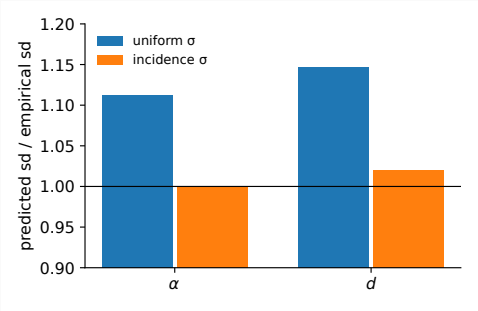
A line with normal $n(\alpha)$ at distance d:

$$r_i = \cos\alpha\, x_i + \sin\alpha\, y_i - d$$

$$\frac{\partial r_i}{\partial(\alpha, d)} = \left[\, -\sin\alpha\, x_i + \cos\alpha\, y_i,\ -1 \,\right]$$

- The first column is a **lever arm**. Points far apart along the wall pin the angle.
- α and d come out correlated ($\rho \approx -0.97$), because d is measured from an origin the points are far away from.

Is the covariance honest?



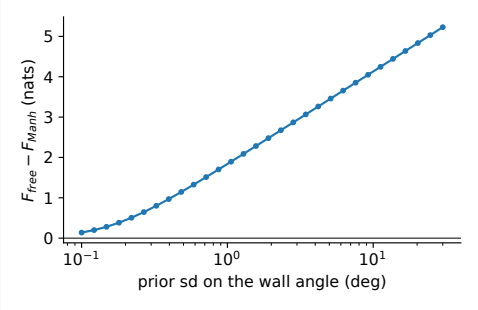
$\Sigma = \mathcal{H}^{-1}$ is a *claim* about how far the fit would move on a re-scan. Test it with 400 scans from one pose:

- uniform σ: about 10% too pessimistic, and it does not average away;
- the noise is along the *beam*, the residual along the wall's *normal*: $\sigma_i = \sigma\,|u_i^\top n|$;
- with that likelihood: within a few per cent.

A Laplace covariance is exactly as honest as the likelihood you wrote.

source: lab 1, exercises 3–4; the real localiser's incidence weight: room_concept.tex l.232

Free energy compares models: the price of a parameter



Two models of the same wall:

- **Manhattan**: the angle is fixed at 90°; one parameter, d.
- **free**: $\alpha \sim \mathcal{N}(90^\circ, s^2)$; two parameters.

Both fit equally well. The free model pays $\approx \ln(s/\sigma_\alpha)$ nats of *complexity*: the cost of squeezing the prior down to the posterior.

Tilt the points by 1.5° and the verdict flips by 70 nats.

Nobody wrote "if the angle error exceeds k degrees". That is the first instance of **no thresholds**.

Why action needs a second functional

Perception is easy to state. Action is not.

Perception. o_t is fixed, so only q can move:

$$q^* = \arg\min_q F[q, o_t]$$

That is approximate Bayesian inference, and nothing else.

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An agent descending the instantaneous free energy finds a flat landscape in every direction.

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What action *can* change is the free energy of observations not yet received:

$$a^* = \arg\min_a \mathbb{E}_{q(o_{t+1}, s_{t+1} \mid a)}\big[F[q, o_{t+1}]\big]$$

This is not a new principle. It changes what is held fixed and what is averaged over.

source: active_inference.tex l.335–369

Expected free energy

For a policy π (a sequence of actions), with the predicted states and observations $\tilde{s}, \tilde{o}$:

$$G(\pi) = \mathbb{E}_{q(\tilde{s},\tilde{o}\,|\,\pi)}\big[\ln q(\tilde{s}\,|\,\pi) - \ln p(\tilde{o},\tilde{s})\big]$$

Goals enter as a **prior over preferred outcomes** $p(\tilde{o})$. There is no separate reward channel. Then, exactly:

$$G(\pi) = \mathbb{E}\Big[\underbrace{\ln q(\tilde{s}\,|\,\pi) - \ln q(\tilde{s}\,|\,\tilde{o},\pi)}_{\text{epistemic}} \underbrace{-\ln p(\tilde{o})}_{\text{pragmatic}} + \underbrace{\ln q(\tilde{s}\,|\,\tilde{o},\pi) - \ln p(\tilde{s}\,|\,\tilde{o})}_{\to 0 \text{ if inference stays good}}\Big]$$

source: active_inference.tex l.371–412

Two terms, one currency

Epistemic value

$= -I(\tilde{s}; \tilde{o} \mid \pi)$: minus the mutual information between future states and future observations.

Largest when what you would see depends strongly on what is true. Curiosity is a *term of the objective*, not a heuristic bolted on.

Pragmatic value

$= -\mathbb{E} \ln p(\tilde{o})$: how improbable the predicted outcomes are under the agent's preferences.

This is goal-seeking: the part a classical cost function would have captured.

Both are in **nats**, and they were derived together. The exploration/exploitation balance is set by the algebra, not by a mixing coefficient. It moves by itself as the beliefs sharpen.

source: active_inference.tex l.414–427

For Gaussian beliefs, three names for one number

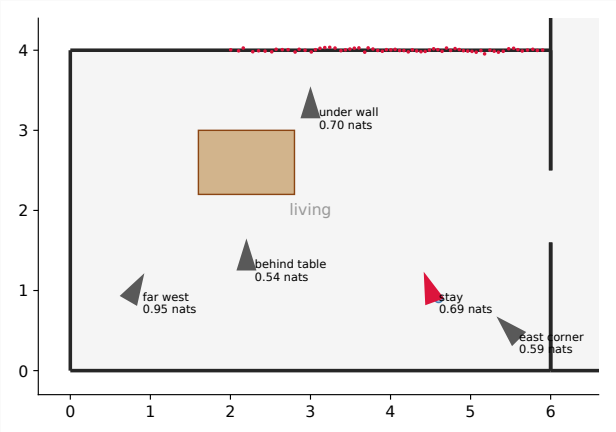
Prior precision Λ, a measurement with Jacobian J and noise R:

$$\underbrace{H[q] - \mathbb{E}H[q \mid o]}_{\text{expected entropy drop}} = \underbrace{I(s; o)}_{\text{mutual information}} = \underbrace{\tfrac{1}{2} \ln \frac{|\Lambda + J^\top R^{-1} J|}{|\Lambda|}}_{\text{D-optimal design}}$$

- The epistemic term of active inference, the information gain of active sensing, and the D-criterion of experiment design are the same quantity.
- It depends on *where* you look (J), not on *what* you will see. So it can be computed before moving.

source: appendix_math.tex l.742–779, l.1236–1287

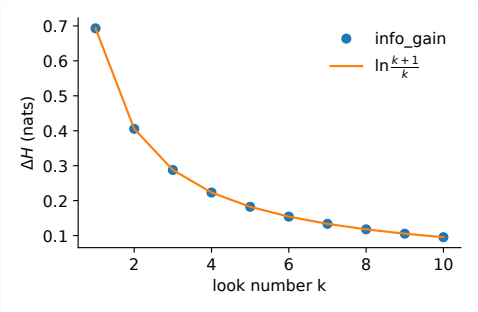
Where to look next



The robot (red) has seen the east part of the north wall. Five candidate viewpoints:

- "far west" sees the *fewest* new points except one, yet gains the *most*: it adds lever arm where the current information is weak;
- "stay" gains exactly $\ln 2$. Why?

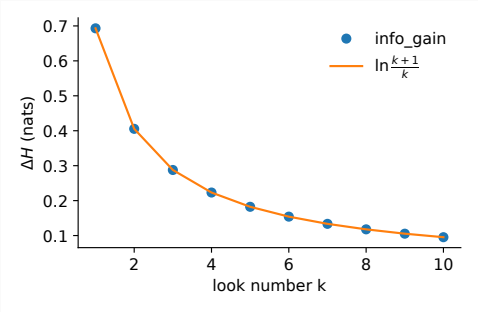
Diminishing returns, and a warning



The k-th identical look gives $|(k{+}1)\Lambda_1|/|\,k\Lambda_1| = ((k{+}1)/k)^2$ for two parameters, so

$$\Delta H_k = \ln \frac{k+1}{k}.$$

Diminishing returns, and a warning



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$$\Delta H_k = \ln \frac{k+1}{k}.$$

But this assumes each look's errors are independent. On a robot, repeated looks share the pose error, the mount error and the calibration error, and those do not average away.

Tomorrow we put a number on how little a repeat look is really worth.

Pricing travel in the same currency

The simplest two-term expected free energy, the one that drives the robot's viewpoint selection:

$$G(v) = \underbrace{\beta \, \|v - x\|}_{\text{cost of travel}} - \underbrace{\Delta H(v)}_{\text{information gain}}$$

- β is an *exchange rate*, in nats per metre, not a tuning knob between two incommensurable objectives.
- In the lab: above $\beta \approx 0.07$ nats/m the robot stays put and scans again, because the (overvalued) $\ln 2$ is free.
- The full version sums agent-owned terms over a horizon. That is Day 5.

source: introduction.tex l.244–253; action.tex

One objective across time scales

The gradient of a T-step expected free energy with respect to the current command splits as

$$\nabla_{u_t} G_T(\pi) = \underbrace{\nabla_{u_t} G_{t+1}(u_t)}_{\text{what a reactive controller computes}} + \underbrace{\sum_{\tau=t+2}^{t+T} \nabla_{u_t} G_\tau(\pi)}_{\text{what planning adds}}$$

- Reactive control is the leading term of planning. They are not two systems.
- The relatives you know (planning-as-inference, MPC, MPPI) sit on this line. They differ in how many terms they keep and how they estimate them.

source: active_inference.tex l.431–510, l.557–579

What measurement said

Is the derived objective better than a good heuristic?

The room explorer, written in nats with no free weights, against the hand-tuned scorer it replaced. Both were run over the **same 594 real room layouts**, paired.

	IoU mean	IoU median	Hausdorff median
hand-tuned scorer	0.9794	0.9880	0.038 m
derived objective	0.9806	0.9870	0.069 m
paired difference	$+0.0013 \pm 0.0030$; better / worse / tied $= 238/298/58$, sign test $p = 0.011$		

- It wins the mean and loses the count. Five rescued rooms carry the whole mean; without them the difference is -0.00085.
- The verdict is equivalence, and the case for the derived form rests on its having no free parameters, not on performance.

On Day 5 we will see that even this IoU was not measuring what we thought it was.

source: experiments.tex l.1404–1446

Free energy is not computation

Measured: a better model does not save arithmetic

Injecting a body-model error from 0 to 10%:

- the solver iterations per correction stay flat to within a twentieth;
- the size of the correction grows by about half as much again.

- The agent pays in *how far its beliefs must move*. The arithmetic doing the moving is nearly constant.
- "A better model saves the agent work" has to say *which* work. The two readings come apart, and an endpoint chosen on the wrong reading measures nothing.

source: active_inference.tex l.835–848

Doing nothing can be an epistemic act

Measured: gyro bias, two legs of the same tour

The share of the bias posterior that came from data rather than from the prior: about **half** on one leg, **almost all** on the other. The only difference was that the robot happened to stand still for a few minutes.

- The bias regresses on elapsed time alone, so the action that teaches it is the one that only spends time: stopping.
- The body is instrumented by everything the robot does, because every action passes through it. The world is sampled only where the robot looks.
- Hence: epistemic actions for rooms and objects, almost none for the body (Day 3).

source: active_inference.tex l.850–880, l.952–968

A factor inherits the errors of what it is estimated through

- The body is learnt from the localiser's corrections, which are fitted against the room. So a wrong map is charged to the wheels.
- A robot in a room whose map is wrong will conclude that its wheels are wrong.
- And it gets *quieter*, not louder: once the body has absorbed the map's distortion, the two agree. The state is internally consistent and externally wrong.

Measured: an omitted parameter

Admitting a camera-mount translation the data cannot support moves the reported pitch by **nine times** its stated interval. An interval speaks only about the parameters the model contains.

Carry an external reference. Never trust only what the agent computes about itself.

source: active_inference.tex l.899–950

Lab

Lab 1: a wall as a belief, and where to look next

`labs/day1.ipynb`, about 75 minutes. Today the robot knows its pose and we cheat on which points are the wall's.

1. Write the wall's residual and Jacobian; fit it with `laplace`.
2. **Honesty test**: 500 scans; predicted vs empirical σ. Find the 10% and fix the likelihood.
3. **Model choice**: Manhattan vs free, by ΔF. Then tilt the wall.
4. **Where to look**: ΔH for five viewpoints. Guess first, then compute.
5. **Diminishing returns**: the k-th look against $\ln \frac{k+1}{k}$.
6. **Price travel**: $G = \beta d - \Delta H$. Find the β at which the robot stops moving. Then answer the day's key question: what does $\ln 2$ assume?
7. Go and look: predicted vs realised entropy drop.

Tomorrow

The robot loses both of today's cheats:

- it no longer knows its pose;
- it no longer knows which points belong to which wall.

One wall becomes a room, the pose becomes one more belief, and the two must be estimated *together*.

And we measure the common-mode ceiling: how little a thousand points can teach when they share one error.

Reading for Day 2

Thesis chapters:

- *Estimating the Layout of an Unknown Room*, §§1–5;
- *The Room Concept Agent*, §§"What is estimated" and "The observation model".